

Daily Science Anecdotes

Topic: The Road That Made Everyone Slower

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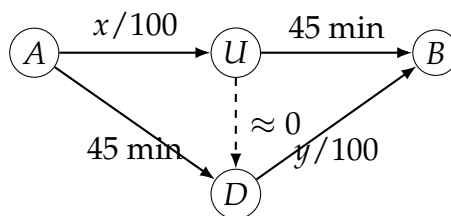
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Imagine a city building a new road to reduce traffic.

The road is faster. The shortcut is almost free to use. Everything about it looks like an improvement.

And yet, after the road is added, the traffic can become *worse for everyone*.

This is not a thought experiment invented for a movie. It is a mathematical phenomenon known as **Braess's Paradox**.



A simplified network illustrating the added shortcut.

In 1968, the German mathematician **Dietrich Braess** studied how traffic distributes itself when each driver chooses what appears to be the best route.

Suppose the two original routes carry flows x and y , with

$$x + y = 4000.$$

Before the shortcut is added, symmetry gives

$$x = y = 2000.$$

If the congested part of a route takes $x/100$ minutes while the other part takes 45 minutes, both routes then require

$$\frac{2000}{100} + 45 = 65 \text{ minutes.}$$

Now add the almost instantaneous connection between the middle points.

A driver can use the first congestible section, take the shortcut, and then use the second congestible section. If all drivers make the same individually attractive choice, the two congestible sections can each carry the full flow:

$$\frac{4000}{100} + \frac{4000}{100} = 80 \text{ minutes.}$$

Thus the equilibrium can change from

$$\boxed{65 \text{ min}} \longrightarrow \boxed{80 \text{ min}}.$$

The new road did not remove congestion.

It changed the *incentives*.

The important distinction is between an individual's best response and the best outcome for the whole system. In game-theoretic language, the resulting state can be a **Nash equilibrium**: no single driver can improve their own travel time by changing route alone. But such an equilibrium need not be socially optimal.

This idea was revisited in modern transportation research. In 2008, **Hyejin Youn, Michael T. Gastner, and Hawoong Jeong** analyzed transportation networks of several major cities and quantified this mismatch using the **price of anarchy**. Their simulations showed that uncoordinated drivers can waste travel time and, counterintuitively, that blocking selected streets can sometimes improve traffic conditions.

So perhaps the strange lesson is not really about roads.

A system can contain many individually reasonable decisions and still produce a collectively inefficient result.

What is best for each individual need not be best for the system.

Sometimes, mathematics becomes most interesting when the "obvious" improvement is exactly what creates the problem.